2-ISOMORPHIC MORPHISMS OF ALGEBRAIC SPACES ARE EQUAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original lemma 04M4: sources/stacks-chow/tags.txt, source lines 6030–6030. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04IN: sources/stacks-chow/tags.txt, source lines 5909–5909. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00Y1: sources/stacks-chow/tags.txt, source lines 1239–1239. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04J3: sources/stacks-chow/tags.txt, source lines 5924–5924. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04LW: sources/stacks-chow/tags.txt, source lines 6022–6022. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let X, Y be algebraic spaces over S. Any two morphisms $a, b : X \to Y$ of algebraic spaces over S for which there exists a 2-isomorphism $(a_{small}, a^\sharp) \cong (b_{small}, b^\sharp)$ in the 2-category of ringed topoi are equal.*

Proof. Let $t : a_{small}^{-1} \to b_{small}^{-1}$ be the 2-isomorphism. We may equivalently think of t as a transformation $t : a_{spaces,étale}^{-1} \to b_{spaces,étale}^{-1}$ since there is not difference between sheaves on $X_{étale}$ and sheaves on $X_{spaces,étale}$. Choose a commutative

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diagram

$$\begin{array}{ccc} U & \xrightarrow[\alpha]{} & V \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ X & \xrightarrow{a} & Y \end{array}$$

where U and V are schemes, and p and q are surjective étale. Consider the diagram

$$\begin{array}{ccc} h_U & \xrightarrow[\alpha]{} & a^{-1}_{spaces,étale} h_V \\ \| & & \downarrow{\scriptstyle t} \\ h_U & \dashrightarrow & b^{-1}_{spaces,étale} h_V \end{array}$$

Since the sheaf $b^{-1}_{spaces,étale} h_V$ is isomorphic to $h_{V \times_{Y,b} X}$ we see that the dotted arrow comes from a morphism of schemes $\beta : U \to V$ fitting into a commutative diagram

$$\begin{array}{ccc} U & \xrightarrow[\beta]{} & V \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ X & \xrightarrow{b} & Y \end{array}$$

We claim that there exists a sequence of 2-isomorphisms

$$\begin{aligned} (\alpha_{small}, \alpha^\sharp) &\cong (\alpha_{spaces,étale}, \alpha^\sharp) \\ &\cong (a_{spaces,étale,c}, a^\sharp_c) \\ &\cong (b_{spaces,étale,d}, b^\sharp_d) \\ &\cong (\beta_{spaces,étale}, \beta^\sharp) \\ &\cong (\beta_{small}, \beta^\sharp) \end{aligned}$$

The first and the last 2-isomorphisms come from the identifications between sheaves on $U_{spaces,étale}$ and sheaves on $U_{étale}$ and similarly for V. The second and fourth 2-isomorphisms are those of Lemma 04M4 with $c : U \to X \times_{a,Y} V$ induced by α and $d : U \to X \times_{b,Y} V$ induced by β. The middle 2-isomorphism comes from the transformation t. Namely, the functor $a^{-1}_{spaces,étale,c}$ corresponds to the functor

$$(\mathcal{H} \to h_V) \longmapsto (a^{-1}_{spaces,étale} \mathcal{H} \times_{a^{-1}_{spaces,étale} h_V, \alpha} h_U \to h_U)$$

and similarly for $b^{-1}_{spaces,étale,d}$, see Sites, Lemma 04IN. This uses the identification of sheaves on $Y_{spaces,étale}/V$ as arrows $(\mathcal{H} \to h_V)$ in $Sh(Y_{spaces,étale})$ and similarly for U/X, see Sites, Lemma 00Y1. Via this identification the structure sheaf $\mathcal{O}_V$ corresponds to the pair $(\mathcal{O}_Y \times h_V \to h_V)$ and similarly for $\mathcal{O}_U$, see Modules on Sites, Lemma 04J3. Since t switches α and β we see that t induces an isomorphism

$$t : a^{-1}_{spaces,étale} \mathcal{H} \times_{a^{-1}_{spaces,étale} h_V, \alpha} h_U \longrightarrow b^{-1}_{spaces,étale} \mathcal{H} \times_{b^{-1}_{spaces,étale} h_V, \beta} h_U$$

over h_U functorially in $(\mathcal{H} \to h_V)$. Also, t is compatible with $a^\sharp_c$ and $b^\sharp_d$ as t is compatible with $a^\sharp$ and $b^\sharp$ by our description of the structure sheaves $\mathcal{O}_U$ and $\mathcal{O}_V$ above. Hence, the morphisms of ringed topoi $(\alpha_{small}, \alpha^\sharp)$ and $(\beta_{small}, \beta^\sharp)$ are 2-isomorphic. By Étale Cohomology, Lemma 04LW we conclude $\alpha = \beta$! Since $p : U \to X$ is a surjection of sheaves it follows that $a = b$. $\square$